Analysis Phase — AI-Based Exam Proctoring System

1. Project Overview

The project is an **AI-Based Exam Proctoring System** designed to provide a controlled and secure environment for conducting university examinations.

The system consists of two main user-facing applications:

1. Flutter Mobile Application: Used by teachers to create and manage exams, define questions and model answers, schedule exams, and view student grades. Students use the mobile application to view upcoming exams, exam schedules, and their grades.

2. Dedicated Exam Application: Used by students to actually take examinations. This application provides identity verification, webcam-based monitoring, and AI-based proctoring during the examination.

The system also includes a backend responsible for managing users, exams, questions, schedules, submissions, grades, and proctoring information.

The mobile application examination is performed through the dedicated exam environment to provide the required monitoring and examination controls.

2. Problem & Proposed Solution

Traditional online examinations face challenges in student identity verification, exam monitoring, cheating prevention, and result management.

The proposed system is an **AI-Based Exam Proctoring System** that provides a controlled examination environment. It consists of a **Flutter mobile application** for exam management and result viewing, and a dedicated exam application for conducting monitored examinations.

Teachers can create and manage exams, schedules, questions, and results through the mobile application. Students can view upcoming exams and their grades through the **mobile application**, while the actual examination is conducted through the dedicated exam application with identity verification and AI-based proctoring.

3. Project Objectives

1. Develop a centralized platform for university exam management.
2. Provide teachers with tools to create, schedule, and manage exams.
3. Allow students to view exam schedules and grades through the mobile application.
4. Provide a dedicated and controlled environment for conducting exams.
5. Verify student identity before the examination.
6. Apply AI-based proctoring to detect predefined suspicious behaviors.
7. Automate exam grading where applicable.
8. Protect student, exam, and result data through appropriate security controls.
9. Generate exam questions automatically from lecture content using AI.
10. Reduce cheating through a secure exam environment and randomized exams.
11. Provide teachers with an integrity score and a collusion report to support review decisions.

4. Stakeholders

Stakeholder	Role in the System
Teacher	Creates and manages examinations, questions, schedules, model answers, and views results.
Student	Views exam information and grades through the mobile application and takes examinations through the dedicated exam application.
AI Proctoring Module	Monitors predefined student behaviors and generates warnings.
System Administrator	Manages system configuration, users, and system-level operations.
Development Team	Manages system configuration, users, and system-level operations.

5. Functional Requirements

Teacher — Mobile Application & Exam Application

FR-01: Login securely.
FR-02: Create and manage exams.
FR-03: Add questions and model answers.
FR-04: Set exam date, time, and duration.
FR-05: Publishes the exam.
FR-06: View student results and grades.
FR-07: Generate exam questions automatically using AI.
FR-08: Upload lecture files (PDF or Word) as the source for question generation.
FR-09: Select the question difficulty level (Easy / Medium / Hard) and the number of questions to generate.
FR-10: Review, edit, or reject AI-generated questions before adding them to the exam.
FR-11: View the Integrity / Suspicion Score of each student after the exam.
FR-12: View the Anti-Collusion report after the exam.

Student — Mobile Application

FR-13: Login securely.
FR-14: View upcoming exams and schedules.
FR-15: View previous grades.
FR-16: Access exam information without taking the exam through the mobile app.

Student — Exam Application

FR-17: Access assigned exams.
FR-18: Verify student identity.
FR-19: Complete webcam and permission requirements.
FR-20: Take and submit exams within the assigned time.
FR-21: Receive warnings for detected suspicious behavior.
FR-22: Terminate the exam when the warning limit is reached.
FR-23: Store answers and calculate grades.
FR-24: Take the exam inside a secure lockdown environment.
FR-25: Receive the exam with randomized question order and answer options.

AI Proctoring

FR-26: Monitor the student during the exam.
FR-27: Detect predefined suspicious behaviors.
FR-28: Record and manage proctoring warnings.
FR-29: Detect suspicious voice or speech through the microphone.
FR-30: Calculate an Integrity / Suspicion Score for each student based on weighted violations.
FR-31: Compare student answers after the exam and detect abnormal similarity patterns, especially in essay questions.

Secure Exam Environment (Details of FR-24)

FR-24.1: Prevent leaving the exam application or switching windows/tabs.
FR-24.2: Block copy/paste, screenshots, and screen recording.
FR-24.3: Log any violation attempt as a proctoring event.

6. Non-Functional Requirements

Security : The system should protect student, examination, answer, and result data from unauthorized access or modification.

Privacy : The system should handle student identity and webcam-related information securely and only for the purposes required by the examination system.

Performance : The system should process examination operations and proctoring events with minimal delay.

Usability : The mobile application and examination application should provide a simple and understandable interface for teachers and students.

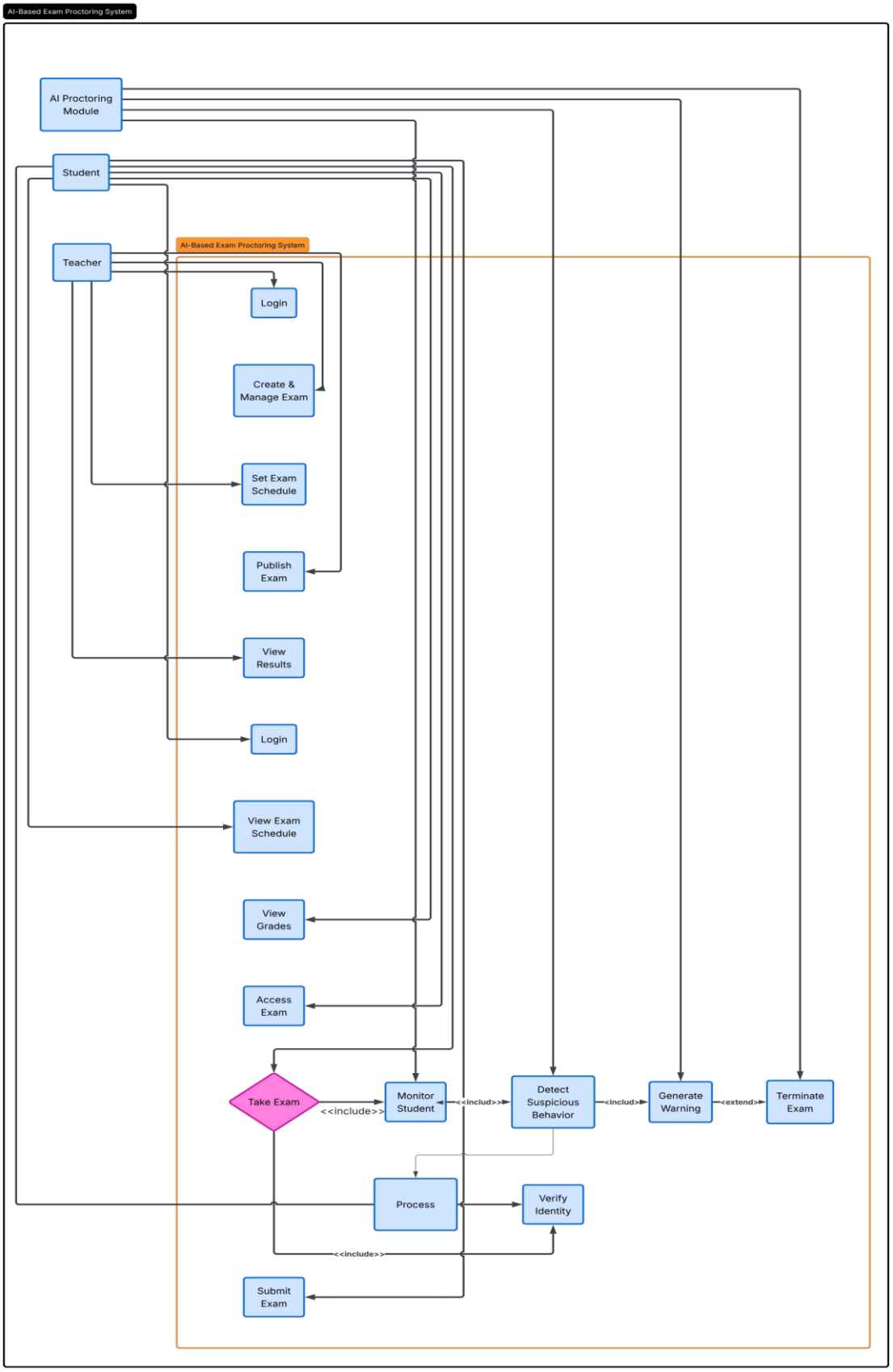
Reliability : The system should maintain examination data and student answers reliably during an examination.

Availability : The system should provide access to examination management and examination services according to the configured examination schedule.

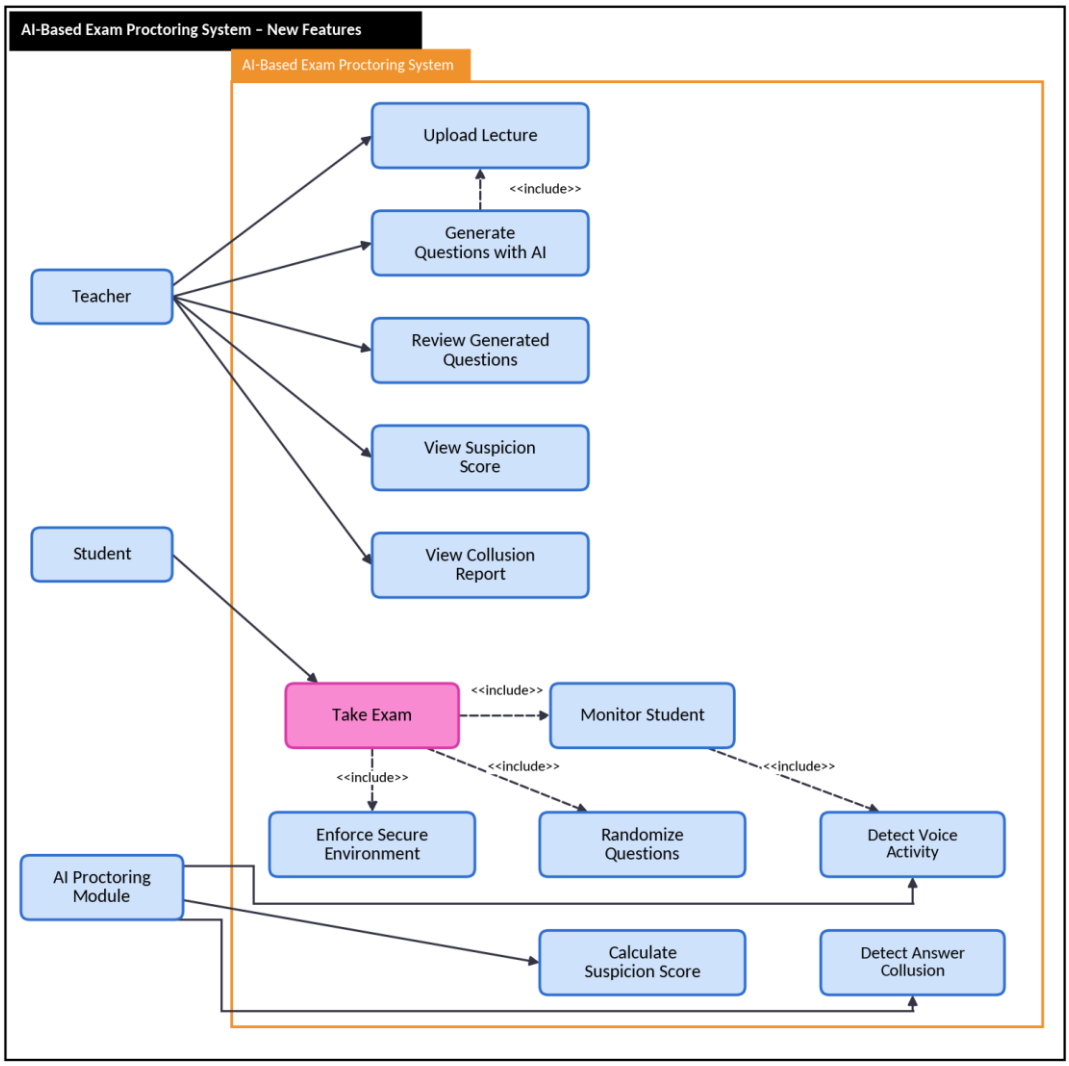
Scalability : The system should be designed so that additional students, examinations, and teachers can be supported as the system grows.

Maintainability : The system should be structured into separate components so that individual components can be maintained and updated without unnecessarily affecting the entire system.

7. Use Case Diagram



7.1 Use Case Diagram — New Features



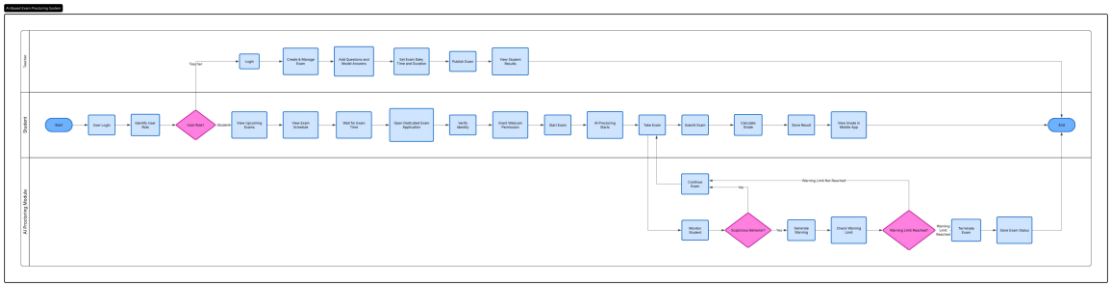
The diagram above shows the use cases added for the new features. They belong to the same AI-Based Exam Proctoring System boundary: Take Exam now includes Enforce Secure Environment and Randomize Questions, and Monitor Student includes Detect Voice Activity.

8. Use Case Descriptions

Use Case	Actor	Description
Login	Teacher, Student	Allows users to securely access the system according to their role.
Create & Manage Exam	Teacher	Allows the teacher to create exams, add questions, and manage exam content.
Set Exam Schedule	Teacher	Allows the teacher to define the exam date, time, and duration.
Publish Exam	Teacher	Makes the exam available to students according to its schedule.
View Results	Teacher	Allows the teacher to view student submissions and grades.
View Exam Schedule	Student	Allows students to view upcoming exams and their scheduled times.
View Grades	Student	Allows students to view their previous exam grades.
Access Exam	Student	Allows an authorized student to access an assigned examination.
Verify Identity	Student	Verifies the student's identity before the examination starts.
Take Exam	Student	Allows the student to answer and submit questions within the assigned time.
Submit Exam	Student	Submits the student's answers for processing and grading.
Monitor Student	AI Proctoring Module	Monitors the student during the examination.
Detect Suspicious Behavior	AI Proctoring Module	Detects predefined behaviors that may violate examination rules.
Generate Warning	AI Proctoring Module	Records and issues a warning when suspicious behavior is detected.
Terminate Exam	AI Proctoring Module	Ends the examination when the configured warning limit is reached.

Use Case	Actor	Description
Upload Lecture	Teacher	Allows the teacher to upload lecture files (PDF or Word) and select the difficulty level and number of questions.
Generate Questions with AI	Teacher	Allows the teacher to generate exam questions automatically from the uploaded lecture using AI.
Review Generated Questions	Teacher	Allows the teacher to review, edit, or reject AI-generated questions before adding them to the exam.
Enforce Secure Environment	Student (Exam Application)	Locks the exam environment by preventing window switching, copy/paste, screenshots, screen recording, virtual machines, and remote access.
Randomize Questions	Student (Exam Application)	Presents each student with a different order of questions and answer options.
Detect Voice Activity	AI Proctoring Module	Detects suspicious speech or sounds through the microphone and generates a warning.
Calculate Suspicion Score	AI Proctoring Module	Calculates an integrity score for each student based on weighted violations recorded during the exam.
View Suspicion Score	Teacher	Allows the teacher to view each student's suspicion score after the exam.
Detect Answer Collusion	AI Proctoring Module	After the exam ends, compares students' answers and detects abnormal similarity patterns, especially in essay questions.
View Collusion Report	Teacher	Allows the teacher to view the list of students flagged for suspicious answer similarity.

9. System Workflow / Activity Diagram



9.1 Workflow Additions for the New Features

1. Teacher: upload lecture, select difficulty and number of questions, AI generates questions, teacher reviews/edits/rejects, approved questions are added to the exam.
2. Before the exam starts: the system randomizes the question order and answer options for each student.
3. During the exam: the secure environment blocks window switching, copy/paste, screenshots, screen recording, virtual machines, and remote desktop tools, and the AI module also monitors the microphone for suspicious voice or speech. Each violation is recorded with a weight.
4. After submission: the system calculates the Integrity / Suspicion Score and runs the Anti-Collusion comparison on student answers; the teacher then views both reports.

10. Data Requirements

The system will manage and store the following main data:

- **User Data:** User ID, name, email, role, and student ID.
- **Exam Data:** Exam ID, title, date, start time, duration, and status.
- **Question Data:** Question ID, exam ID, question text, options, model answer, and marks.
- **Submission Data:** Student answers, submission time, score, and status.
- **Proctoring Data:** Warning events, event type, timestamp, warning count, and exam status; voice event type, security violation type (window switch, copy/paste, VM detected, etc.), and violation weight.
- **Schedule Data:** Exam date, time, duration, and availability status.
- **Lecture Data:** Lecture ID, teacher ID, file name, file type (PDF/Word), upload date, and extracted text.
- **Generated Question Data:** Question ID, lecture ID, difficulty level, question text, options, suggested answer, and review status (Pending / Approved / Rejected).
- **Randomization Data:** Student ID, exam ID, question order, and options order.
- **Suspicion Score Data:** Student ID, exam ID, total score, list of violations with weights, and risk level (Low / Medium / High).
- **Collusion Data:** Exam ID, student pair, question ID, similarity percentage, and flagged status.

11. Project Constraints & Assumptions

Constraints

1. The project is developed within the available university project timeline.
2. The system is limited by the hardware available for development and testing.
3. AI detection accuracy may depend on camera quality, lighting, and environmental conditions.

4. Webcam-based monitoring requires the student to have a functioning camera.
5. The system's performance may depend on the available computing resources.
6. The project team has limited time and resources for implementing all possible examination features.
7. The system may require specific device or operating-system requirements for the dedicated examination environment.
8. Automated grading may be limited depending on the question types supported by the system.
9. AI-generated questions depend on the quality and clarity of the uploaded lecture content.
10. Voice detection accuracy may depend on microphone quality and background noise.
11. Full lockdown features may depend on the operating system of the student's device.
12. Anti-collusion detection is less accurate on short or objective answers.

Assumptions

1. Teachers have valid accounts and authorization to create examinations.
2. Students have valid university accounts.
3. Students provide correct identification information.
4. Students have access to a functioning device capable of running the examination application.
5. The required camera is available and functional during the examination.
6. Students provide the permissions required by the examination application.
7. Teachers define appropriate model answers where automated grading is used.
8. The AI module is configured to detect the predefined suspicious behaviors supported by the project.
9. The university provides the required rules and policies for conducting examinations through the system.
10. Students have a functioning microphone during the exam.
11. Teachers review AI-generated questions before publishing the exam.
12. The Suspicion Score and Collusion Report are decision-support tools, and the final decision is made by the teacher or the university.

12. Development Plan

Phase 1 – System & Web/Mobile Foundation

Design the system architecture and database.
Develop user authentication.
Develop the teacher interface.
Allow teachers to create and manage exams.

Phase 2 – Exam Management

Add questions and model answers.
Set exam date, time, and duration.
Allow students to view upcoming exams and schedules.
Upload lectures (PDF or Word) and select difficulty and number of questions.
Implement AI Question Generation.
Review and edit AI-generated questions.

Phase 3 – Dedicated Exam Application

Develop the student exam environment.
Implement identity verification.
Implement webcam permissions and exam access.
Implement the Secure Exam Environment (lockdown mode).
Implement Dynamic Question Randomization.

Phase 4 – AI Proctoring

Implement suspicious behavior detection.
Generate warnings.
Terminate the exam when the warning limit is reached.
Implement Voice / Speech Detection.
Implement the Integrity / Suspicion Score.
Implement Anti-Collusion Detection (after the exam).

Phase 5 – Testing & Integration

Integrate the mobile/web system, exam application, backend, and AI module.
Test security, functionality, performance, and exam workflow.
Test the lockdown environment on different operating systems.
Test AI question quality, voice detection accuracy, and collusion detection accuracy.